

Supplementary Online Material for: Conditional modal branch selection in physics-informed active control of lid-driven cavity flow

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This document provides the implementation details, complete datasets, and additional analyses supporting the main manuscript. Section numbers S1–S11 correspond to the references “Supplementary Sec. S...” used in the text.

1 S1. PINN architecture and training details

The parametric network represents the flow-control pair for the full Reynolds interval $[100, 1000]$ through inputs $(x, y, t, \widetilde{Re})$, with $\widetilde{Re} = (Re - 550)/450 \in [-1, 1]$.

- **Flow subnetwork:** 6 hidden layers $\times$ 64 neurons, tanh activation, 21 379 trainable parameters; outputs (u, v, p) .
- **Control subnetwork:** 4 hidden layers $\times$ 32 neurons, 3 614 trainable parameters; outputs the coefficients $\{a_{mn}\}$.
- **Precision:** all arithmetic in double precision; residuals by automatic differentiation.
- **Training:** 300-epoch pre-training (flow only, control frozen at zero), then 3 000-epoch joint phase with Adam ($\eta_0 = 8 \times 10^{-4}$, decay factor 0.995 per 100 epochs). Collocation points: $N_{\text{phys}} = 4\,000$ interior points resampled uniformly every epoch; $N_{\text{probe}} = 20$ for the Reynolds-separation term.
- **Loss weights:** $\lambda_{\text{PDE}} = 1$, $\lambda_{\text{BC}} = 15$, $\lambda_{\text{diss}} = 0.05$, $\lambda_{\text{ctrl}} = 200$, $\lambda_{\text{var}} = 1$, $\lambda_{Re} = 2$.

2 S2. Modal-energy quadrature conventions

For the lid-expansion $U_{\text{lid}}(x, t, Re) = \sum_m \sum_n a_{mn} \sin(m\pi x/L_x) \cos(n\pi t)$, the modal energy accounting uses $\langle \cos^2(0) \rangle = 1$ and $\langle \cos^2(n\pi t) \rangle = \frac{1}{2}$ for $n \geq 1$:

$$\mathcal{E}_{m0} = \frac{1}{2} a_{m0}^2, \quad \mathcal{E}_{mn} = \frac{1}{4} a_{mn}^2 \quad (n \geq 1).$$

The realized actuation energy is $E_r = \langle U_{\text{lid}}^2 \rangle_{s,t}$, with $s = x/L_x$, and the normalized modal fractions $f_{mn} = \mathcal{E}_{mn} / \sum \mathcal{E}_{ij}$ sum to unity. The stationary second-harmonic fraction is $f_2 \equiv f_{20}$ and the total time-dependent fraction is $f_t = \sum_m \sum_{n \geq 1} f_{mn}$.

Table 1: Pooled and per-condition statistics of the 185-realization campaign (30 cells; oversampled cells: $Re = 1000, E^* = 1$ ($n = 20$), $Re = 500, E^* = 0.25$ ($n = 15$), $Re = 700, E^* = 2$ ($n = 15$)).

Quantity	B_1	B_2	B_{first}	B_3
Pooled frequency (frozen labels)	42.2%	43.8%	0.0%	14.1%
Pooled frequency (threshold re-classification)	42.2%	43.8%	0.0%	14.1%
Mean per condition	42.1%	46.7%	0.0%	11.3%
Pooled excluding oversampled cells ($N = 135$)	42.2%	48.1%	0.0%	9.6%
Discrepancies frozen vs. threshold ($N = 185$)	0 / 185			

3 S3. Reference (Re, E^*) campaign – realization-level data

The threshold classification rule is: B_{first} if $f_1 > 0.5$; B_1 if $f_2 > 2f_t$ and $f_2 > 0.5$; B_2 if $f_t > 2f_2$ and $f_t > 0.5$; otherwise B_3 . No realization of the 185-run reference campaign at $\lambda_{\text{var}} = 1$ satisfies $f_1 > 0.5$, so B_{first} is empty there. The frozen labels and the threshold-based re-classification coincide for all 185 realizations (0/185). The realization-level file (Supplementary Table S1 in the submission system) lists, for each realization: (Re, E^*), seed, f_2 , f_t , the dominant coefficients, total loss, and frozen branch label.

4 S4. Variance-regularization campaign and Wilson intervals

Six values of $\lambda_{\text{var}} \in \{0, 0.25, 0.5, 1, 2, 5\}$ at three operating points, $N_s = 5$ seeds each (90 runs). Frequencies are reported in Table 3 of the main text. Finite-sample 95% Wilson-score confidence intervals use

$$p \pm \frac{z}{\sqrt{N_s}} \sqrt{\frac{\hat{p}(1 - \hat{p})}{N_s} + \frac{z^2}{4N_s^2}} \bigg/ \left(1 + \frac{z^2}{N_s}\right), \quad z = 1.96.$$

For $N_s = 5$ and $\hat{p} = 0, 0.2, 0.4, 0.6, 1.0$ the half-widths are 0.31, 0.43, 0.47, 0.43, 0.31, respectively. All realizations are classified with the four-class rule of Sec. S3. At $\lambda_{\text{var}} = 0$ every one of the 15 realizations is B_{first} (first harmonic, $f_1 > 0.5$); these were grouped with B_3 under the original three-class scheme. The corrected per-cell frequencies and entropies (normalised by $\ln 4$) are those of Table 3 of the main text.

5 S5. Energy sweep and seed multiplicity

Energy sweep at $Re = 500$ ($E^* \in \{0.01, 0.05, 0.1, 0.25, 0.5, 1, 2\}$):

Table 2: Energy sweep at $Re = 500$.

E^*	0.01	0.05	0.1	0.25	0.5	1	2
f_2 (%)	~ 94	~ 93	92.2	92.2	–	–	74.0
f_t (%)	~ 0.5	~ 0.7	1.1	1.3	–	–	21.8

In every energy-sweep realization the first-harmonic fraction remains small ($f_1 \leq 8\%$), so the B_{first} branch never appears; the sweep stays within the B_1 branch and the erosion of second-harmonic purity at high energy is a redistribution toward time-dependent modes rather than toward the fundamental.

Seed multiplicity at $Re = 500$, $E^* = 0.25$, 10 seeds: 4 B_1 , 4 B_2 , 2 B_3 (0 B_{first} , $f_1 < 0.02$ for all seeds); total losses in $[1.371, 1.394]$ (relative spread 1.7%).

6 S6. Capacity and truncation experiments

Capacity variants at $Re = 500$, $E^* = 0.25$: 4×32 (small), 6×64 (reference), 8×128 (large) give $f_2 \in [89.5\%, 92.2\%]$. Control truncations $(N_x, N_t) \in \{(4, 3), (6, 1), (6, 3), (6, 5), (8, 5), (8, 9)\}$ give $f_2 \in [87.3\%, 92.3\%]$ with $A_2 \in [0.668, 0.684]$; see Table 6 of the main text.

7 S7. Spectral bases and aspect ratio

Fourier basis: sine in space (enforces corner compatibility) and cosine in time ($n = 0 = \text{stationary}$). Modified-Chebyshev basis: same structure with Chebyshev polynomials of the first kind T_m on the lid interval, normalized to the same energy budget. Aspect-ratio cells $AR \in \{0.5, 1, 2\}$ at $Re = 500$, $E^* = 0.25$: $f_2 = 69.9\%$, 92.2% , 91.2% (see Table 7 of the main text).

8 S8. MRT-LBM solver and grid-convergence index (GCI)

Solver. D2Q9 lattice with the diagonal relaxation vector $\mathbf{S} = \text{diag}(0, 1.4, 1.4, 0, 1.2, 0, 1.2, \omega, \omega)$, $\omega = 1/\tau$, $\tau = 3\nu_{\text{LBM}} + 0.5$, $\nu_{\text{LBM}} = U_{\text{ref}}N/Re$ with $U_{\text{ref}} = 0.05$. Lid forcing through a velocity boundary condition; boundary populations via the Zou–He scheme. Parallelism via Numba **prange** over lattice rows.

Benchmark. Centerline velocity errors against Ghia et al. at $N = 256$: $L_2(u) = 0.00506$, $L_\infty(u) = 0.01283$ ($Re = 100$); $L_2(u) = 0.01411$, $L_\infty(u) = 0.03003$ ($Re = 1000$); corresponding v -component errors 0.00675/0.01136 and 0.01321/0.02125.

GCI. Three-grid study of the $A_2 \sin(2\pi x)$ control at $Re = 500$, $N \in \{128, 256, 512\}$:

$$\varepsilon_{128} = 0.0093500, \quad \varepsilon_{256} = 0.0104070, \quad \varepsilon_{512} = 0.0109658.$$

Observed order $p_{\text{obs}} = 0.9196$ (from $p = \ln |(\varepsilon_{256} - \varepsilon_{128})/(\varepsilon_{512} - \varepsilon_{256})| / \ln 2$), Richardson extrapolation $\varepsilon_\infty = 0.01159$. The GCI (formula S8.4) uses the safety factor $F_s = 1.25$ and refinement ratio $r = 2$:

$$\text{GCI}_j = F_s \frac{|\varepsilon_{j+1} - \varepsilon_j|}{(r^p - 1)\varepsilon_j}, \quad \Rightarrow \quad \text{GCI}_{512} = 7.15\%, \quad \text{GCI}_{256} = 14.25\%.$$

Uniform-to-extrapolated reduction: $0.01159/0.0442824 = 73.8\%$ (vs. 76.5% measured at $N = 256$).

9 S9. Symbolic regression: configuration and LOCO scores

PySR over binary operators $\{+, -, \times, /\}$ and unary $\{\sin, \cos, \exp, \text{square}\}$, 20 populations, 100 iterations each, Pareto selection on complexity–accuracy trade-off. Best expression: $U_{\text{SR}} \approx a[b + \cos(\omega_{\text{SR}}x - \phi)]^2(c_0 + c_1x + c_2t + c_3Re) + c_4$, $\omega_{\text{SR}} \approx 3.42$. Leave-one-Reynolds-condition-out R^2 over $Re \in \{100, 300, 400, 500, 600, 700, 800, 900, 1000\}$: minimum 0.9326 ($Re = 900$), maximum 0.9908 ($Re = 500$); all nine values above 0.93.

Table 3: Reproducibility double-run results (A/B/original).

seed	run	E	f_2	f_t	A_{10}	A_{20}	branch
4	A	0.2514	0.5957	0.3693	0.0376	-0.5472	B_3
4	B	0.2512	0.6080	0.3575	0.0394	-0.5527	B_3
4	orig	0.2495	0.5215	0.4501	0.0364	-0.5101	B_3
7	A	0.2508	0.9271	0.0083	-0.0604	0.6819	B_1
7	B	0.2508	0.9271	0.0083	-0.0604	0.6819	B_1
7	orig	0.2498	0.9131	0.0162	-0.0681	0.6754	B_1
8	A	0.2507	0.0357	0.9609	-0.0109	0.1339	B_2
8	B	0.2507	0.0357	0.9609	-0.0109	0.1339	B_2
8	orig	0.2503	0.0133	0.9841	-0.0158	0.0817	B_2

10 S10. Reproducibility double runs and multi-seed audit

Double runs A/B (idempotent, same seeds/hyperparameters, base 6×5 , $E^* = 0.25$, final code) at $Re = 500$:

Runs A and B are bit-identical to the last decimal for seeds 7 and 8, and quasi-identical for seed 4 (same class B_3). All three coincide with the original realizations re-analyzed with the final code.

Multi-seed audit (99 realizations, final code, 18 configurations):

Table 4: Audit summary per configuration: mean f_2 , mean f_t , mean A_2 (in %), and branch counts.

label	n	$\bar{f}_2$	$\bar{f}_t$	$\bar{A}_2$	$B_1/B_2/B_3$
ar_rect_2	5	90.8	1.2	0.687	5/0/0
ar_rect_half	5	15.3	83.8	0.187	1/4/0
ar_square	5	73.0	20.1	0.572	4/1/0
bas_cheb_re100	5	0.5	99.2	-0.030	0/5/0
bas_cheb_re1000	5	1.2	98.4	-0.047	0/5/0
bas_cheb_re500	5	2.1	97.6	-0.085	0/5/0
bas_fourier_re100	5	32.6	64.7	0.313	1/3/1
bas_fourier_re1000	5	73.5	21.3	0.610	3/0/2
bas_fourier_re500	5	73.0	20.1	0.572	4/1/0
cap_baseline	8	64.0	30.5	0.402	5/2/1
cap_large	8	89.1	3.6	0.676	8/0/0
cap_small	8	92.2	1.3	0.694	8/0/0
mc_4x3	5	54.4	39.4	0.436	3/2/0
mc_6x1	5	91.1	0.0	0.683	5/0/0
mc_6x3	5	75.1	16.1	0.485	4/1/0
mc_6x5	5	73.1	20.1	0.572	4/1/0
mc_8x5	5	59.1	34.9	0.485	3/2/0
mc_8x9	5	47.4	47.7	0.408	2/2/1

Every audited realization reproduces the frozen branch label of the original run; combined with the 0/185 discrepancies of the threshold re-classification (Sec. S3), the branch taxonomy is internally consistent.

11 S11. Energy-matched comparison, trajectory tests, and B2 windows

Energy-matched LBM at $E^* = 0.25$, $Re = 500$ (integrated dissipation ε and ratio to the uniform reference):

Table 5: Iso-energetic dissipation ratios at $N = 256$ and $N = 512$.

Grid	$\varepsilon_{\text{unif}}$	$\varepsilon_{k=1}$	$\varepsilon_{k=2}$	ratios $k1/k2$
$N = 256$	9.4980×10^{-3}	6.8912×10^{-3}	1.1285×10^{-2}	0.7255 / 1.188
$N = 512$	1.10981×10^{-2}	7.1725×10^{-3}	1.18987×10^{-2}	0.6463 / 1.0721

The energy-matched ranking $k = 1 < \text{uniform} < k = 2$ is unchanged across resolutions; the $k = 1$ ratio range is $[0.646, 0.726]$ and the $k = 2$ (B_1) ratio range is $[1.072, 1.188]$.

Exp D trajectory tests ($Re = 500$, $E^* = 0.25$, probe spacing 50 epochs; seeds 4/7/8 $\leftrightarrow$ classes $B_3/B_1/B_2$). Control-energy autocorrelation $R_E(\Delta)$, control-cycle displacements D_1, D_4 , and modal displacements Δf :

Table 6: Exp D trajectory diagnostics.

seed	$R_E(1)$	$R_E(2)$	$R_E(4)$	D_1	D_4/D_1	$\Delta f(4)$ steady
4	+0.245	+0.167	+0.364	0.0744	2.92	0.0500
7	-0.446	+0.488	+0.372	0.0091	0.92	0.0011
8	+0.552	+0.821	+0.643	0.4178	2.29	0.0010

The four-epoch spectral peak of the raw per-epoch loss ($f \approx 0.25$; power up to $\sim 55\times$ the spectral median for seed 4) does not propagate to energy, control, or modal fractions: no period-4 cycle is detected in the control path, and branch labels stay stable across transients (seed 7: case A; seed 8: finite relocation transient within one class, case B; seed 4: slow persistent drift, case C). With $n = 60$ trajectory points ($n = 30$ stationary), the autocorrelation standard error is $\approx 0.13\text{--}0.18$.

B2 long-window quasi-stationarity ($Re = 500$, $E^* = 0.25$, $N = 256$; LBM run of 640 000 iterations, actuation period 256):

Table 7: B2 kinetic-energy fluctuation ratio over T and $2T$ windows.

window	cycles	$R_K = K_{\text{fluct}}/K_{\text{mean}}$	$ \Delta R_K /R_K(2T)$
T	128	14.95941409258121	—
$2T$	256	14.959414094980596	1.60×10^{-10}

with a 15% persistence threshold, the verdict is **PERSISTANT**: the B_2 flow is statistically stationary in the mean-dissipation sense while being strongly time-dependent at the actuation frequency ($R_K \simeq 15$).